

Product Design Spec - smoke-all

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Features

Feature	Priority	Status	Description
BR-8 — Bound of this feature. **F-3** does not re-validate grammar, does not decide arity, does not format diagnostics beyond the error-handling rules below, and does not define exit codes for arity errors	medium	planned	**BR-8** — Bound of this feature. **F-3** does not re-validate grammar, does not decide arity, does not format diagnostics beyond the error-handling rules below, and does not define exit codes for arity errors
EC-14 — Digit-limit guard	medium	planned	**EC-14** — Digit-limit guard
API-3 — Function-level contract	medium	planned	**API-3** — Function-level contract
none — F-3 is the core value-producing feature of the tool; nothing in it is optional for a correct `add`.	**none** — F-3 is the core value-producing feature of the tool; nothing in it is optional for a correct `add`.		
F-5 local ID scope	medium	planned	**AC-1**–**AC-14**, **BR-1**–**BR-7**, **V-1**–**V-8**, **EC-1**–**EC-14**, **EH-1**–**EH-6**, **API-1**–**API-8** are scoped to this feature section and may be reused with different meanings in other feature sections.
B-5 Behaviour is identical whether the program is invoked as `python3 add.py 2 3`, as an installed console script, or via any other launcher the packaging adopting agent chooses, provided the same interpreter and the same arguments are used. The three-way contract	medium	planned	**B-5** Behaviour is identical whether the program is invoked as `python3 add.py 2 3`, as an installed console script, or via any other launcher the packaging adopting agent chooses, provided the same interpreter and the same arguments are used. The three-way contract
The interpreter's own integer-string conversion ceiling	medium	planned	The interpreter's own integer-string conversion ceiling
BR-7	medium	planned	**The interpreter's integer→string digit cap is a host artifact outside this tool's contract; for otherwise-valid input it MUST NOT be surfaced to the user as an error.**
BR-2	medium	planned	CLI tests MUST invoke the program as a subprocess using `sys.executable` and the project's CLI entry point, so tests do not depend on `PATH`, virtual environments, or installed packages.
BR-5	medium	planned	Test cases MUST be grouped by feature area and named descriptively (e.g., `test\_arity\_rejects\_zero\_operands`, `test\_grammar\_accepts\_leading\_zeros`, `test\_sum\_mixed\_signs`, `test\_render\_negative`, `test\_exit\_status\_success`, `test\_diagnostic\_arity`, `test\_usage\_output`).
A failing test MUST report the test name, the expected value, and the actual value for stdout, stderr, or exit status as applicable.	medium	planned	A failing test MUST report the test name, the expected value, and the actual value for stdout, stderr, or exit status as applicable.
AC-2	medium	planned	Every functional requirement in F-1 through F-10 has at least one automated test case that exercises its externally observable behaviour.
Feature priority	medium	planned	**must-have** — the product plan explicitly includes unit tests for the `add` function and CLI, and the tests are the primary automated verification of all contracts in F-1 through F-10.
AC-2	medium	planned	No option/flag parsing is performed; a leading ` ` is treated as a sign character, not a flag.

Source stages: design